Functional Specification Document (FSD)

ServiceNow Incident Lifecycle Automation

Phase: Project Documentation

1. Document Control

Field	Details
Project	ServiceNow Incident Lifecycle Automation
Document owner	
Version	1.0
Date	
Status	Draft for review
Reviewers	

2. Introduction

2.1 Purpose

This document describes what the automated Incident Management solution does from the user and business point of view. It is the reference for design, development, testing and acceptance.

2.2 Project background

The solution manages an IT incident from creation to closure using automated workflows, Business Rules and Flow Designer in ServiceNow. It integrates Knowledge Management and Change Management to speed up resolution, reduce recurring incidents and improve IT service efficiency.

2.3 Scope

- In scope: incident creation, priority, assignment, state control, notifications, SLAs, auto-close, Knowledge and Change integration, Problem linkage, reporting.
- Out of scope: other ServiceNow applications, hardware and network changes, third-party monitoring tools unless listed in section 8.

2.4 Definitions

Term	Meaning
Incident	An unplanned interruption or reduction in quality of an IT service
SLA	Service Level Agreement: agreed response and resolution times
MTTR	Mean time to resolve
Business Rule	Server-side script that runs when a record is inserted, updated, deleted or queried

Term	Meaning
Flow Designer	ServiceNow low-code tool for building automated workflows
KB	Knowledge Base article
Change Request	A controlled request to change the IT environment

3. System Overview

An incident is raised by a user through the portal, email, phone or an integration. The system sets priority, assigns the incident to a group and notifies the people involved. The agent works on the incident with Knowledge suggestions, and may raise a Change Request or link a Problem. On resolution the system captures the fix, offers to create a Knowledge article and closes the incident after confirmation or a set period.

3.1 User roles

Role	Description
Caller	Reports an incident and confirms resolution
Service Desk agent	Logs, triages and resolves incidents
Resolver group member	Diagnoses and fixes assigned incidents
Incident manager	Oversees SLAs, escalations and major incidents
Knowledge manager	Reviews and publishes articles
Change manager	Approves linked changes
Administrator	Maintains configuration, rules and flows

4. Functional Requirements

ID	Function	Description	Priority
FR-01	Incident creation	Create incidents from portal, email, phone and integration; enforce mandatory fields (caller, short description, category, impact, urgency)	High
FR-02	Priority calculation	Calculate priority from impact and urgency using the agreed matrix; recalculate when either changes	High
FR-03	Auto-assignment	Assign the incident to the group defined for the category, subcategory and configuration item	High
FR-04	State control	Support New, In Progress, On Hold, Resolved, Closed and Canceled with allowed transitions only	High
FR-05	Hold validation	Require a hold reason to set On Hold; pause the SLA while on hold	High
FR-06	Resolution validation	Require resolution code and notes to set Resolved	High

ID	Function	Description	Priority
FR-07	Notifications	Notify caller and group on creation, assignment, state change and resolution; escalate P1 and P2 to the manager	High
FR-08	SLA management	Start, pause and stop SLA timers; warn at 75% elapsed; escalate on breach	High
FR-09	Auto-close	Close Resolved incidents after the agreed period unless the caller reopens	Medium
FR-10	Duplicate detection	Flag likely duplicates by caller, configuration item and description	Medium
FR-11	Knowledge integration	Suggest articles, allow attaching, and offer article creation on resolution	Medium
FR-12	Change integration	Link or raise a Change Request; put the incident On Hold until the change closes	Medium
FR-13	Problem linkage	Link recurring incidents to a Problem record	Medium
FR-14	Reporting	Provide reports and dashboards for volume, MTTR, SLA compliance, backlog and recurring incidents	Medium

5. Functional Specifications

5.1 State model

From	To (allowed)	Condition
New	In Progress, On Hold, Canceled	Assignment complete for In Progress
In Progress	On Hold, Resolved, Canceled	Resolution fields complete for Resolved
On Hold	In Progress	Hold reason cleared or linked change closed
Resolved	Closed, In Progress	Caller confirms, period expires, or caller rejects (back to In Progress)
Closed / Canceled	None	Terminal states; record is read-only

5.2 Priority matrix

Impact / Urgency	High urgency	Medium urgency	Low urgency
High impact	P1 Critical	P2 High	P3 Moderate
Medium impact	P2 High	P3 Moderate	P4 Low
Low impact	P3 Moderate	P4 Low	P5 Planning

Confirm the matrix with the business before build; the table above is the proposed standard.

5.3 Business Rules

ID	Name	When	Logic
BR-01	Calculate priority	Before insert / update	Set priority from impact and urgency
BR-02	Require hold reason	Before update	Abort if state is On Hold and hold reason is empty
BR-03	Require resolution info	Before update	Abort if state is Resolved and resolution code or notes are empty
BR-04	Set auto-close date	After update	When state becomes Resolved, set the close date from the agreed period
BR-05	Duplicate check	After insert	Match caller, configuration item and short description; flag possible duplicate

5.4 Flows

ID	Flow	Trigger	Main actions
FL-01	Incident intake	Incident created	Assign group; notify caller and group; escalate if P1 or P2
FL-02	SLA monitoring	SLA at 75% and at breach	Notify assignee and manager; escalate on breach
FL-03	Resolution and close	State is Resolved	Send confirmation request; close after period; prompt for Knowledge article
FL-04	Change hold	Change linked to incident	Set On Hold; release when the change closes

5.5 Notifications

Event	Recipients	Content
Incident created	Caller, assignment group	Number, short description, priority
Assigned	Assignee	Incident details and SLA target
State changed	Caller	New state and next step
Resolved	Caller	Resolution summary and request to confirm
SLA warning / breach	Assignee, manager	Incident, time remaining or breached

5.6 Knowledge and Change integration

- Knowledge: suggestions use category and short description; an attached article is recorded on the incident; a draft article can be created from a resolved incident and goes to the Knowledge manager for review.

- Change: a Change Request can be raised from the incident; the incident is On Hold with the change as the reason and is released when the change closes; the caller is then asked to confirm.
- Problem: recurring incidents on the same configuration item or category can be linked to a Problem, whose permanent fix follows Change Management.

6. Data Specification

Field	Type	Notes
Number	String	Auto-generated unique identifier
Caller	Reference (User)	Mandatory
Short description	String	Mandatory
Category / Subcategory	Choice	Mandatory; drives assignment
Impact / Urgency	Choice	Mandatory; drive priority
Priority	Choice	Calculated; read-only to agents
Assignment group / Assigned to	Reference	Set by flow; editable by agents
State	Choice	Controlled by state model
Hold reason	Choice	Required when On Hold
Resolution code / notes	Choice / Text	Required when Resolved
Linked Knowledge article	Reference	Optional
Linked Change Request / Problem	Reference	Optional

7. Non-functional Requirements

Area	Requirement
Performance	Incident form loads and saves in a few seconds under normal load; targets agreed in performance testing
Security	Role-based access; only permitted roles can change priority, close or cancel; sensitive data restricted
Auditability	State changes, assignments and resolutions are recorded in the activity history
Usability	Mandatory fields marked; clear messages when a validation blocks an action
Maintainability	Prefer configuration and Flow Designer over scripting; changes moved by update sets
Availability	Follows the ServiceNow platform availability; failures in integrations do not stop incident creation

8. Interfaces and Integrations

Interface	Direction	Notes
Knowledge Management	Internal	Suggest, attach and create articles
Change Management	Internal	Link and raise Change Requests
Email	Inbound / outbound	Create incidents and send notifications
Other systems (if any)	Inbound / outbound as applicable	Document evidence during implementation/review

9. Assumptions and Constraints

- Categories, groups, SLA targets and the priority matrix are confirmed by the business before build.
- Knowledge Management and Change Management are active on the instance.
- The solution is developed and tested on a non-production instance before deployment.
- Targets and values in this document are proposed and subject to approval.

10. Traceability to Testing

Requirement	UAT scenarios	Performance testing area
FR-01, FR-02, FR-03	UAT-01, UAT-02	Incident form and flows
FR-05, FR-06	UAT-03, UAT-04	Business Rules
FR-08, FR-09	UAT-05, UAT-06, UAT-07	SLA and flows
FR-11	UAT-08, UAT-09	Gen AI, AI and ML features
FR-12, FR-13	UAT-10, UAT-11, UAT-12	Salesforce integration (if used)
FR-14	UAT-13	Power BI and Tableau dashboards

11. Approval

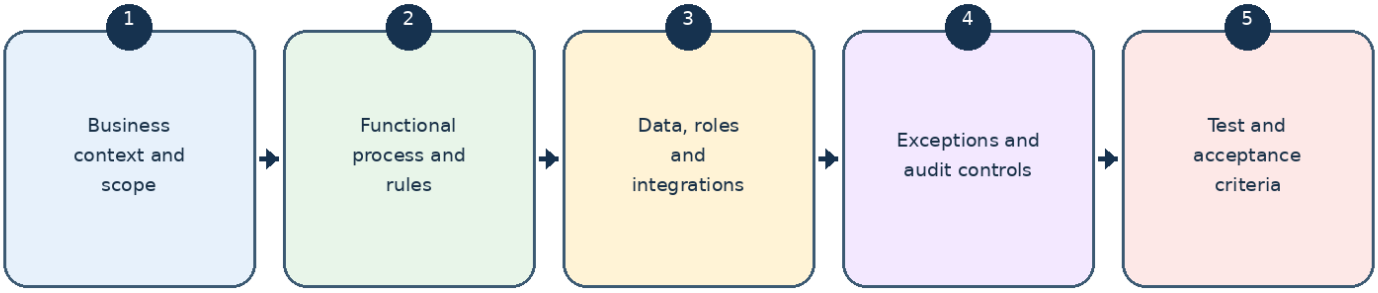
Name	Role	Signature	Date
Incident Lifecycle Automation	Business owner		To be confirmed by project team
Incident Lifecycle Automation	Project owner		To be confirmed by project team

Name	Role	Signature	Date
Incident Lifecycle Automation	Reviewer		To be confirmed by project team

Process Diagram

FSD Documentation

Incident Lifecycle Automation • Process diagram



Expected output: a traceable, testable process with clear ownership and documented outcomes.